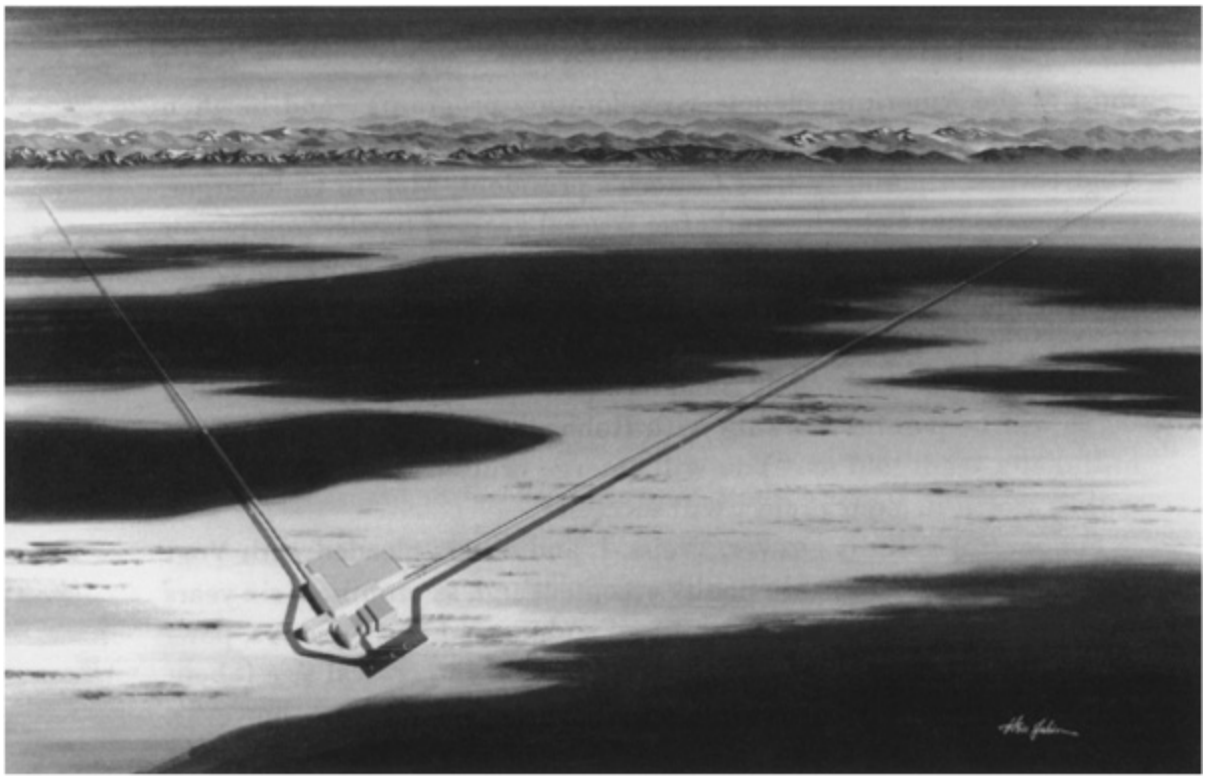
A key to success in our endeavor will be the construction and operation of a national scientific facility called the *Laser Interferometer Gravitational-Wave Observatory,* or *LIGO.* The LIGO will consist of two L-shaped vacuum systems, one near Hanford, Washington, and the other near Livingston, Louisiana, in which physicists will develop and operate many successive generations of ever-improving interferometers; see Figure 10.8.

Why two facilities instead of one? Because Earth-bound gravitational-wave detectors always have ill-understood noise that simulates gravitational-wave bursts; for example, the wire that suspends a mass can creak slightly for no apparent reason, thereby shaking the mass and simulating the tidal force of a wave. However, such noise almost never happens simultaneously in two independent detectors, far apart. Thus, to be sure that an apparent signal is due to gravitational waves rather than noise, one must verify that it occurs in two such detectors. With only one detector, gravitational waves cannot be detected and monitored.



10.8 Artist's conception of LIGO's L-shaped vacuum system and the experimental facilities at the corner of the L, near Hanford, Washington. [Courtesy LIGO Project, California Institute of Technology.]